



Problem Statement

Optimizing RAG Document Retrieval for Agronomic Advice

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Smallholder farmers across Sub-Saharan Africa often face urgent, specific problems—a nutrient deficiency, a pest outbreak, a disease on their crop—but the expert knowledge to solve these problems is locked away in scattered agricultural extension material that is hard to search effectively. Simple keyword-based tools often fail here: a search for “yellow maize leaves” can just as easily return a document about potassium deficiency when the real cause is nitrogen deficiency, because the wording looks similar even though the answer is wrong. These *hard negatives* are exactly what separates a system that truly understands a farmer’s question from one that only matches surface-level words.

This project builds the retrieval engine at the core of a Retrieval-Augmented Generation (RAG) system for agricultural extension advice. Given a farmer’s question, the system ranks a knowledge base of **695 factsheets** covering crop diseases, pests, nutrient deficiencies, soil management, climate adaptation, and fertiliser advice, returning the **top-5 most relevant documents**. Because relevance is graded from a perfect answer down to a related-but-wrong document—ranking order matters, not just whether a relevant document appears somewhere in the results.

We move beyond the provided TF-IDF baseline (~ 0.55 nDCG@5) by exploring:

- **BM25** for improved lexical retrieval,
- **Dense sentence-embedding retrieval** for semantic matching, and
- **Cross-encoder re-ranking** for fine-grained relevance discrimination.

The goal is to better distinguish true relevance from superficial keyword overlap, particularly on hard negatives where lexical methods are known to fail. Performance is measured using **nDCG@5**, which rewards placing the most relevant, expert-verified documents at the very top of the ranking not just anywhere in the list.

By improving retrieval quality in this way, the project moves closer to a system that can reliably surface the correct expert guidance for a farmer’s exact problem—turning existing agricultural knowledge into something genuinely usable at the moment it is needed, rather than something that merely looks related.